

Def: An **almost complex structure** on M is a smooth family

$$J_x: T_x X \longrightarrow T_x X$$

with $J_x^2 = -1$.

If M is symplectic, we say that J is **compatible** if $\omega(\cdot, J\cdot)$ is a Riemannian metric.

Note that any symplectic manifold has a compatible almost complex structure.

Let

$$TM \otimes \mathbb{C}$$

$$\downarrow$$

$$M$$

be the (fibrewise) complexification of the tangent bundle. Extend J to $TM \otimes \mathbb{C}$ by

$$J(v \otimes c) = Jv \otimes c.$$

Then J has eigenvalues $\pm i$. Let

$$T_{1,0} = \{v \in TM \otimes \mathbb{C} : Jv = iv\} = \{v \otimes 1 - Jv \otimes i : v \in TM\},$$

$$T_{0,1} = \{v \in TM \otimes \mathbb{C} : Jv = -iv\} = \{v \otimes 1 + Jv \otimes i : v \in TM\},$$

$$\pi_{1,0}: TM \xrightarrow{\cong} T_{0,1}, v \mapsto \frac{1}{2}(v \otimes 1 - Jv \otimes i),$$

$$\pi_{1,0}: TM \xrightarrow{\cong} T_{1,0}, v \mapsto \frac{1}{2}(v \otimes 1 + Jv \otimes i).$$

For the cotangent bundle, we have

$$T^{1,0} = \{\eta \in T^*M \otimes \mathbb{C} : \eta(J\cdot) = i\eta(\cdot)\} \longleftarrow T^*M, \frac{1}{2}(\eta - i\eta \circ J) \longleftarrow \eta,$$

$$T^{0,1} = \{\eta \in T^*M \otimes \mathbb{C} : \eta(J\cdot) = -i\eta(\cdot)\}.$$

Let

$$\Lambda^{l,m} = \Lambda^l T^{1,0} \wedge \Lambda^m T^{0,1},$$

so that $\Lambda^k = \bigoplus_{l+m=k} \Lambda^{l,m}$, where

$$\Lambda^k = \Lambda^k(T^*M \otimes \mathbb{C}).$$

Let $\Omega^{l,m} = \Gamma(X, \Lambda^{l,m})$ with projections $\pi^{l,m}: \Omega^{l+m} \longrightarrow \Omega^{l,m}$. Let

$$\partial = \pi^{l+1,m} \circ d: \Omega^{l,m} \longrightarrow \Omega^{l+1,m}$$

$$\bar{\partial} = \pi^{l,m+1} \circ d: \Omega^{l,m} \longrightarrow \Omega^{l,m+1}.$$

Note that on Ω^k ,

$$d = \sum_{r+s=k+1} \pi^{r,s}.$$

We say that $f: X \longrightarrow \mathbb{C}$ is **(J)-holomorphic** if $\partial f \circ J = i\partial f$, ie if $\partial f \in T^{1,0}$, and antiholomorphic otherwise, ie if $\bar{\partial} f$ is holomorphic.

Note that f is holomorphic iff $\bar{\partial} f = 0$, and antiholomorphic iff $\partial f = 0$. Note also that on functions,

$$d = \partial + \bar{\partial}.$$

Now assume that M has a complex structure. Then locally, we can choose coordinates on TM $\{\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}, \dots, \frac{\partial}{\partial x_n}\}$ such that

$$J\frac{\partial}{\partial x_i} = \frac{\partial}{\partial y_i}, J\frac{\partial}{\partial y_i} = -\frac{\partial}{\partial x_i}.$$

Let

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right), \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right),$$

so that

$$T_{1,0} = \text{span}_{\mathbb{C}} \left\{ \frac{\partial}{\partial z_j} \right\}, T_{1,0} = \text{span}_{\mathbb{C}} \left\{ \frac{\partial}{\partial \bar{z}_j} \right\}.$$

Similarly,

$$\partial z_j = \partial x_j + i\partial y_j, \partial \bar{z}_j = \partial x_j - i\partial y_j,$$

These are the differentials of a holomorphic chart only if the almost complex structure is integrable!

so that

$$T^{1,0} = \text{span}_{\mathbb{C}} \{ \partial z_j \}, T^{0,1} = \text{span}_{\mathbb{C}} \{ \partial \bar{z}_j \}.$$

Note that for forms of any degree,

$$d = \partial + \bar{\partial}.$$

Thm: Let (M, J) be an almost complex manifold. Then M is a complex manifold iff $d = \partial + \bar{\partial}$ iff $\bar{\partial}^2 = 0$. In this case,

$$H_{\text{Dol}}^{l,m}(M) = H_{\bar{\partial}}^{l,m} = H^m(M, \underline{\Omega}^{(l,0)}).$$

We say that a symplectic manifold (M, ω) is **Kähler** if it has a compatible complex structure. Note that then

$$g(\cdot, \cdot) := \omega(\cdot, J\cdot)$$

is symmetric, so that $J^*\omega = \omega$ and $\omega \in \Omega^{2,0}$.

Let

$$\omega = \frac{i}{2} \sum h_{j\bar{k}} \partial z_j \wedge \partial \bar{z}_k.$$

Since $\bar{\omega} = \omega$, $(h_{j\bar{k}})$ is Hermitian.

strictly plurisubharmonic

Def: Let M be a complex manifold. Then $\varphi \in C^\infty(M; \mathbb{R})$ is **spsh** if the matrix $\left(\frac{\partial^2 \varphi}{\partial \bar{z}_j \partial z_k} \right)$ is positive definite.

Prop: If φ is spsh then $\omega := \frac{i}{2} \partial \bar{\partial} \varphi$ is Kähler. The converse is true locally.

this follows from a calculation (only $\partial \bar{\omega} = 0$ is not immediate).

this follows from the local triviality of H_{Dol} .

(or: complex submanifolds of Kähler manifolds are Kähler.

Thm (Hodge): $H_{\mathbb{R}}^k(M) \cong \bigoplus_{l+m=k} H_{\text{Dol}}^{l,m}(M).$